Chapter 3 Electricity

Lesson 1 Electric Energy

Objectives You will be able to:

-Calculate the electrical energy, power, energy cost.

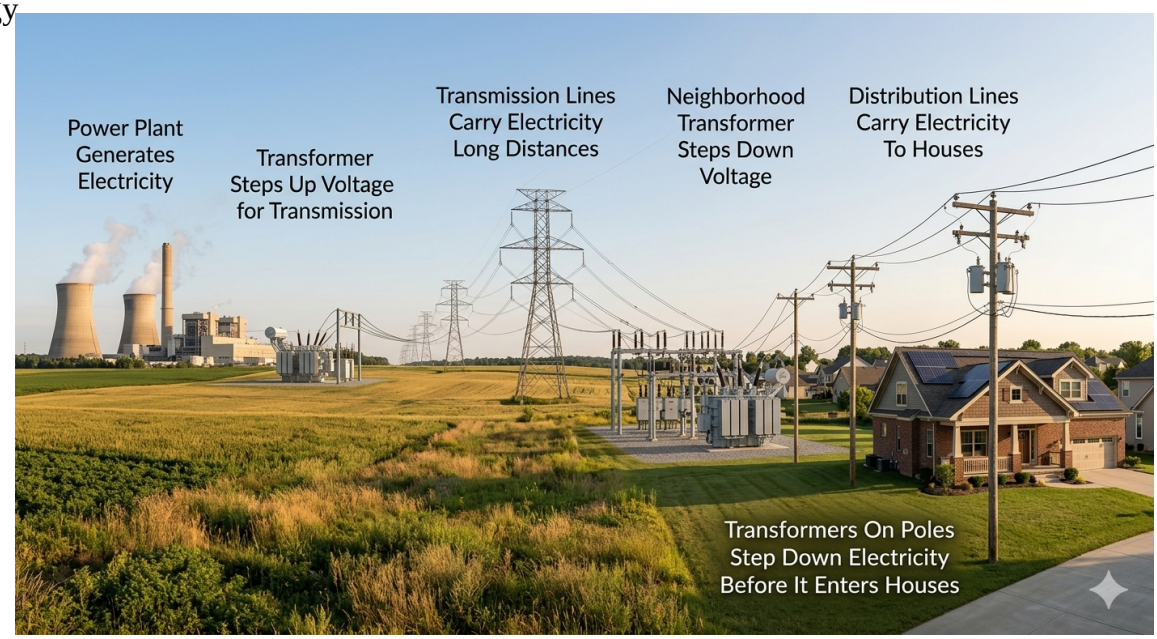
-Understand the change from electrical energy to heat energy.

-Understand how electrical energy transmitted.

1-Electricity generation



Electricity generation is the process of generating electric energy from other forms of energy



Sources of Electrical energy

The energy resources fall into two main categories, often called renewable and non-renewable energy resources. Each of these resources can be used as a source to generate electricity, which is a very useful way of transferring energy from one place to another such as to the home or to industry. Non-renewable sources of energy include fossil fuels and nuclear fuel, natural gas. Renewable energy includes hydro power plant, wind mill, solar energy.

2-Electrical Energy

A heater with a power of 1watt (W) changes 1 joule of electrical energy into heat energy every second. With 1 joule of energy, you could run a 1watt heater for 1 second. With 6 joules of energy, you could run a 2watts heater for 3 seconds.

To calculate energy in joules, use the equation:

E	=	**P**	×	**t**
(joules)		(watts)		(seconds)
(J)		(W)		(s)

The units on an electricity bill are units of energy called **kilowatt hours (kWh)**. The electricity board changes you a set amount for each kWh bought.

With 1 kWh of energy, you could run a 1kilowatt heater for 1 hour.

With 8 kWh of energy, you could run a 1kilowatt heater for 8 hours, or a 2 kilowatt heater for 4 hours. To calculate energy in kilowatt hours, we use the equation:

Energy	=	**power**	×	**time**
(kWh)		(kilowatts)		(hours)
E	=	**P**	×	**t**
(kWh)		(kW)		(h)

3-The cost of using electrical Energy

Electric meter is used to measure the electrical energy used by electrical appliances. Electric meter is measured in kilowatt hours (kWh).

The cost of operating an electrical device can be calculated using the equation:

$$Cost = E(kWh) \times price\ per\ 1kWh$$

Practice Problems

1. A 6-V battery delivers 0.5A of current to an electric motor connected across its terminals.
   a. What is the power rating of the motor?
   b. How much energy does the motor use in 5.0 minutes?
2. The current through the starter motor of a car connected to a 12-V battery is 210A.What electric energy is delivered to the starter in 10s?
3. A flashlight is connected across a 3V difference in potential. The current through the lamp is 1.5A.
   a. What is the power rating of the lamp?
   b. How much electric energy does the lamp convert in 10 min?
4. A lamp draws a 0.50A from a 120-V generator.
   a. How much power does the generator deliver?
   b. How much energy does the lamp convert in 5 minutes?
5. A 15-Ω electric heater operates on a 120-V outlet.
   a. What is the current through the heater?
   b. How much energy is used by the heater in 30.0 seconds?
   c. How much heat is liberated by the heater in this time?
6. A 30-Ω resistor is connected to a 60-V battery.
   a. What is the current in the circuit?
   b. How much energy is used by the resistor in 5 minutes?
7. A 100-W light bulb is 20% efficient. That means 20% of the electric energy is converted to light energy.
   a. How many joules does the light bulb convert into light each minute it is in operation?
   b. How many joules of heat does the light bulb produce each minute?
8. How much energy does a 60.0-W light bulb use in half an hour? If the light bulb is 12% efficient, how much heat does it generate during the half hour?
9. The new color television set draws 2.0A when operated on 120V.
   a. How much power does the set use?
   b. If the set is operated for an average of 7.0 hours per day, what energy in kWh does it consume per month (30days)?
   c. At $0.08 per kWh, what is the cost of operating the set per month?

10. An electric space heater draws 15.0A from a 120-V source. It is operated,

on the average, for 5.0h each day.

a. How much power does the heater use?

b. How much energy in kWh does it consume in 30 days?

c. At $0.08 per kWh, what does it cost to operate the heater in 30 days?

11. A digital clock has an operating resistance of 12 000 Ω and is plugged into a 115-V outlet.

a. How much current does it draw?

b. How much power does it use?

c. If the owner of the clock pays $0.09 per kWh, what does it cost to operate the clock for 30 days?

4-Electrical Energy and Heat Energy

4-1-Heat Energy

Heat is the transfer of thermal energy between substances. Thermal energy is the kinetic energy of moving particles of matter, measured by their temperature. Heat always transfers from a warmer substance to a colder substance.

Temperature is measured in °C, and heat is measured in J. When heat energy is transferred to an object, its temperature increases depends upon the:

-The mass of the object.

-The substance the object is made from.

-The amount energy transferred to the object.



For a particular object, the more heat energy transferred to it, the greater its temperature increase.

This relationship is known as Joule's first law.

4-2-Specific Heat Capacity

The specific heat capacity of a substance is the amount of heat energy needed to change the temperature of 1 kg of the substance by 1°C. Different substances have different specific heat capacities.

Notice that water has a particularly high specific heat capacity. This makes water useful for storing heat energy, and for transporting it around the home using central heating pipes.

The table below shows the specific heat of some common substances

Specific heat of common substance			
Material	Specific heat	Material	Specific heat

	(J/kg°C)		(J/kg°C)
Alcohol(ethanol)	2450	Iron	450
Aluminum	903	Lead	130
Brass	376	Silver	235
Carbon	710	Steam	2020
Copper	385	Water	4190
Glass	664	Zinc	388
Gold	129	Ice	2060

4-3-Calculating heat gained or lost

The temperature raises 1°C of 1-kg water required heat Energy of 4190J.

The temperature raises 1°C of 2-kg water required heat Energy of 8380J.

The temperature raises 10°C of 2-kg water required heat Energy of 83800J.

☞E is the energy transferred in joules, J

☞m is the mass of the substances in kg

☞c is the specific heat capacity in J / kg °C

☞ Δt is the temperature change in degrees Celsius, °C

According to the above statement, heat gained or lost is defined by:

$$\text{Heat Energy} = \text{Mass} \times \text{Specific heat capacity} \times \text{Temperature rise}$$

$$Q = m \times c \times \Delta t$$

(J) (kg) (J/kg°C) (°C)

For example, how much energy must be transferred to raise the temperature of 2 kg of water from 20°C to 30°C?

E = m × c × θ (θ = 30 – 20 = 10°C)

E = 2 × 4181 × 10 = 83,620 J or 83.62 kJ

Practice Problems

1. How much heat is absorbed by $2.50\times10^2\,g$ of water when it's heated from 10°C to 85°C?

2. How much heat in kJ is needed to raise the temperature of 50g of water from 45 °C to 83 °C?

3. How much heat in joules must be added to 50kg of aluminum at 25 °C to raise its temperature to 125 °C?

4. (a) How much heat is required to raise the temperature of 250mL of water from 20°C to 35°C?

(b) How much heat is lost by the water as it cools back down to 20°C?

5. How much heat does 25g of aluminum give off as it cools from 100°C to 20°C? For aluminum $c = 903 J/kg^oC$.

6-(a) Calculate the amount of energy needed to change the temperature of:

(i) 500 g of water by 5°C (specific heat capacity of water is 4200 J/kg°C)

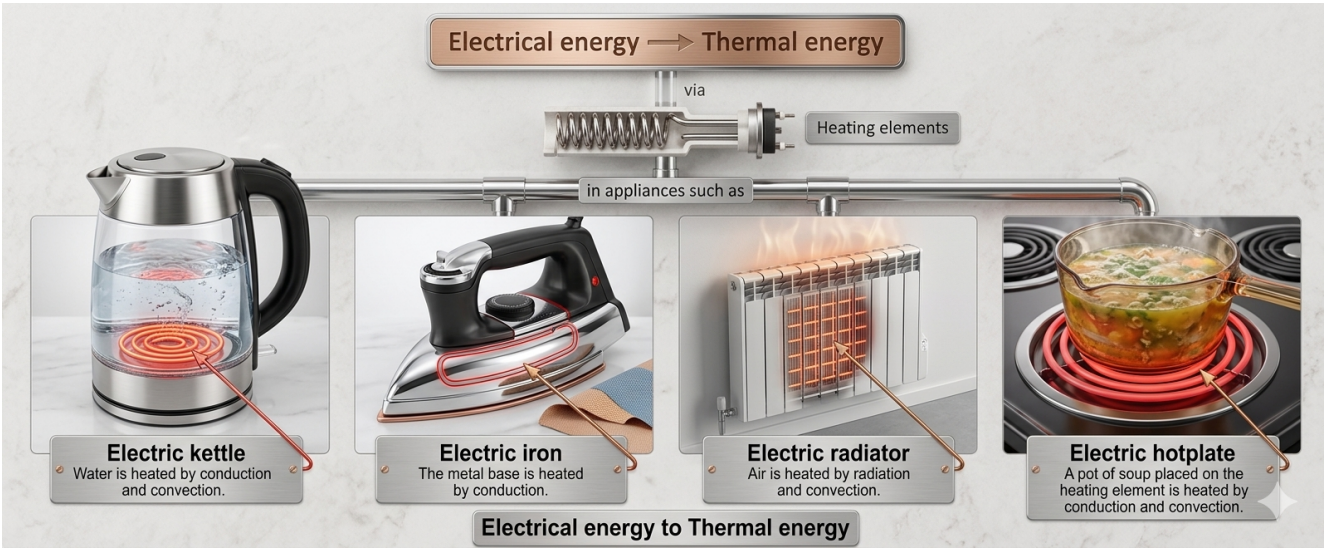
(ii) 100 g of aluminum from 100C to 300C (specific heat capacity of aluminum is 880 J/kg °C).

(b)A 4 kg lump of iron is given 20 kJ of energy. It rises in temperature by 10°C. Calculate the specific heat capacity of iron.

4-4 Electrical energy and heat energy

Many household appliances convert electrical energy to thermal energy. These appliances have heating elements that are usually made of nichrome, an alloy of Nickel and chrome. Nichrome is used because it has high resistivity and can withstand high temperatures. When electric current passes through these elements, they heat up rapidly.

Electric heating is any process in which electrical energy is converted to heat. Common applications include space heating, cooking, water heating and industrial processes. An **electric heater** is an electrical appliance that converts electrical energy into heat. Common appliances include electric kettles, irons, radiators, hotplates. An electric current through the heating element converts electrical energy into heat energy.



Summing no loss of energy to the surrounding: $Q = E$

$$E = R \times I^2 \times t$$

$$Q = m \times c \times \Delta t$$

Where t is time in second

Where t_f, t_i final and initial temperature in (^{0}C)

where Q is the heat generated by a constant current I flowing through a conductor of electrical resistance R, for a time t. When current, resistance and time are expressed in amperes, ohms, and seconds respectively, the unit of E is the joule.

There are many practical uses of Joule heating. Some of the most common are as follows.

- An incandescent light bulb glows when the filament is heated by Joule heating, so hot that it glows white with thermal radiation (also called blackbody radiation).
- Electric stoves and other electric heaters usually work by Joule heating.
- Soldering irons and cartridge heaters are very often heated by Joule heating.
- Electric fuses rely on the fact that if enough current flows, enough heat will be generated to melt the fuse wire.

Practice Problems

1. A heating coil has a resistance of $4\,\Omega$ and operates on 120V.



a. What is the current in the coil while it is operating?

b. What energy is supplied to the coil in 5 min?

c. If the coil is immersed in an insulated container holding 20kg of water, what will be the increase in the temperature of the water? Assume that 100% of the heat is absorbed by the water.

2-An electric heater is rated at 1000W.

a. How much energy is delivered to the heater in 2 hours?

b. The heater is being used to heat a room containing 100 kg of air. If the specific heat of air is (1100 J/kg^{o}C) and 70% of the thermal energy heats the air in the room, what is the change in air temperature?

3. The resistance of an electric stove element at operating temperature is $11\text{k}\,\Omega$.

a. 220V is applied across it. What is the current through the element?

b. How mush energy does the element use in 30mins?

c. The element is being used to heat a kettle containing 1.20 kg of water. Assume that 70% of the heat is absorbed by the water. What is its increase in temperature during the 30s?

4. An electric heater is rated at 500W.

a. How much energy is delivered to the heater in half an hour?

b. The heater is being used to heat a room containing 50 kg of air. If the specific heat of air is 1.10 kJ/kg^{o}C (1100 J/kg^{o}C) and 50% of the thermal energy heats the air in the room, what is the change in air temperature?

5-A consumer requires an immersion heater for a tank containing 140liters of

Water. The water is to be heated from 6^0C to 66^0C. 3.5hours. Calculate the nearest element size (in kilowatts) if the heating system is 90 percent efficient.

1 liter weights 1kg. and 1kWh=3600000J.

6-A water heater has a volume of 10litres and is fitted with a 2kW immersion heater. Calculate the time taken to raise the temperature of water from 4^0C to

46^0C. The water heater is 85 percent efficient.

7- A water heater takes 5hours to heat 180litres of water from 5^0C to 60^0C.

Calculate the efficiency of water if the element is 3kW.

8- A current was sent through a resistance wire of 10 ohms, which is fully immersed in 2 kg of water. At the end of 15 minutes, the rise in temperature was observed to be 60^0C. Determine the value of current.

